

Physical and Engineering Sciences in Medicine

Estimation of Organ and Effective Doses of CBCT Scans of Radiotherapy Using Size-Specific Field of View (FOV): A Monte Carlo Study --Manuscript Draft--

Manuscript Number:	APES-D-23-00431
Full Title:	Estimation of Organ and Effective Doses of CBCT Scans of Radiotherapy Using Size-Specific Field of View (FOV): A Monte Carlo Study
Article Type:	Scientific Paper
Keywords:	Monte Carlo simulations Cone beam CT (CBCT) Image-guided radiation therapy (IGRT) Field of view (FOV) Dosimetry
Abstract:	kV cone beam CT (CBCT) is one of the most common imaging modalities used for image-guided radiation therapy (IGRT) procedures. The utilization of CBCT scans during a course of cancer treatment plays a vital role in improving the treatment outcome. However, extra doses are delivered to patients, thus assessment and optimization of the imaging doses should be taken into consideration. Unlike conventional CT, CBCT scans are acquired with pre-set scanning protocols. One of the parameters set by the vendor is the size of the field of view (FOV), which is applied for all patients regardless of body size. This study aimed to investigate the influence of using fixed and patient-specific FOVs on the patient dose. Monte Carlo simulations were performed to simulate kV beams of the Varian OBI imaging system integrated into Truebeam linear accelerator using BEAMnrc code. Organ and size-specific effective doses resulting from chest and pelvis scanning protocols were estimated with DOSXYZnrc code using a phantom library developed by the National Cancer Institute (NCI) of the US. The library contains 193 (100 male and 93 female) mesh-type computational human adult phantoms, and it covers a large ratio of patient sizes with heights and weights ranging from 150 to 190 cm and 40 to 125 kg. The imaging doses were assessed using variable FOV of three sizes, small (S), medium (M), and large (L) for each scan region. The S FOV covered a scan length of 18 cm along the phantom body, while 22 cm and 26 cm were used for the M and L FOVs, respectively. The results show that the FOV and the patient size played a major role in the scan dose. The average percentage differences (PDs) for doses of organs that were fully inside the different FOVs were relatively low, all within 12% for both protocols. However, doses to organs that were scanned partially or near the FOVs were affected significantly. For the chest protocol, inclusion of the thyroid in the scan field could give a dose of 1–7 mGy/100 mAs to the thyroid, compared to 0.4–1 mGy/100 mAs when it was excluded. Similarly, on average, testes doses could be 6 mGy/100 mAs for the male pelvis protocol compared to 3 mGy/100 mAs when it did not lie in the field irradiated. These dose differences resulted in an average increase of up to 27% in the size-specific effective dose of the protocols. Since changing the field size option is unavailable for the OBI system, it can be suggested that various patient-specific scanning protocols are developed for each scan area. The patient's height might be considered as the main factor upon which, the selection of the appropriate patient-specific protocol is based. The patient weight was found to impact the imaging dose; thus, the weight should also be taken into consideration. This approach should optimize the imaging doses by minimizing doses of a large ratio of patients.

Estimation of Organ and Effective Doses of CBCT Scans of Radiotherapy Using Size-Specific Field of View (FOV): A Monte Carlo Study

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Abstract

kV cone beam CT (CBCT) is one of the most common imaging modalities used for image-guided radiation therapy (IGRT) procedures. The utilization of CBCT scans during a course of cancer treatment plays a vital role in improving the treatment outcome. However, extra doses are delivered to patients, thus assessment and optimization of the imaging doses should be taken into consideration. Unlike conventional CT, CBCT scans are acquired with pre-set scanning protocols. One of the parameters set by the vendor is the size of the field of view (FOV), which is applied for all patients regardless of body size. This study aimed to investigate the influence of using fixed and patient-specific FOVs on the patient dose.

Monte Carlo simulations were performed to simulate kV beams of the Varian OBI imaging system integrated into Truebeam linear accelerator using BEAMnrc code. Organ and size-specific effective doses resulting from chest and pelvis scanning protocols were estimated with DOSXYZnrc code using a phantom library developed by the National Cancer Institute (NCI) of the US. The library contains 193 (100 male and 93 female) mesh-type computational human adult phantoms, and it covers a large ratio of patient sizes with heights and weights ranging from 150 to 190 cm and 40 to 125 kg. The imaging doses were assessed using variable FOV of three sizes, small (S), medium (M), and large (L) for each scan region. The S FOV covered a scan length of 18 cm along the phantom body, while 22 cm and 26 cm were used for the M and L FOVs, respectively.

The results show that the FOV and the patient size played a major role in the scan dose. The average percentage differences (PDs) for doses of organs that were fully inside the different FOVs were relatively low, all within 12% for both protocols. However, doses to organs that were scanned partially or near the FOVs were affected significantly. For the chest protocol, inclusion of the thyroid in the scan field could give a dose of 1–7 mGy/100 mAs to the thyroid, compared to 0.4–1 mGy/100 mAs when it was excluded. Similarly, on average, testes doses could be 6 mGy/100 mAs for the male pelvis protocol compared to 3 mGy/100 mAs when it did not lie in the field irradiated. These dose differences resulted in an average increase of up to 27% in the size-specific effective dose of the protocols.

Since changing the field size option is unavailable for the OBI system, it can be suggested that various patient-specific scanning protocols are developed for each scan area. The patient's height might be considered

1 as the main factor upon which, the selection of the appropriate patient-specific protocol is based. The patient
2 weight was found to impact the imaging dose; thus, the weight should also be taken into consideration. This
3 approach should optimize the imaging doses by minimizing doses of a large ratio of patients.

4

1. Introduction

The use of imaging in clinical practice is an essential part, not only in the diagnosis of diseases but also performing radiotherapy treatments. A variety of imaging modalities are used for different treatment procedures. Imaging is considered the main guidance tool for overall cancer treatment stages, starting from the diagnosis through to the follow-up. Cancer treatment has evolved significantly over time, especially in the last two decades. One of those improvements is the image-guided radiation therapy (IGRT) technique, which was introduced in 2000's alongside the development of advanced technology of radiotherapy treatments such as intensity-modulated radiotherapy (IMRT) and stereotactic ablative radiotherapy (SABR) (Jaffray et al., 2002). The IGRT technique assists in the optimization of the treatment by enabling the position of the tumour to be determined accurately immediately prior to delivery of prescribed doses to cancer tumours. Various imaging modalities are used for IGRT, for instance, kV and MV cone beam computed tomography (CBCT), magnetic resonance imaging (MRI), radiography, portal imaging, and fluoroscopy (Murphy et al., 2007; Ding et al., 2018).

A recent international survey has been carried out and concluded that kV-CBCT is the most common imaging modality utilized for IGRT procedures worldwide (C. J. Martin et al., 2021). The CBCT scan is performed by a kV system that is usually integrated into a linear accelerator (LINAC) and operated independently just before the beginning of the radiotherapy treatment (Boda-Heggemann et al., 2011; Posiewnik and Piotrowski, 2019). This allows the positioning of the patient on the LINAC table precisely by comparing images of CBCT (the treatment position) with those acquired by a CT simulator that is used for the treatment planning (the reference position). The accurate position of the patient results in the determination of the target, and hence increases tumour control probability, and reduces the risk of treatment side effects by, for example, minimizing doses to healthy tissues. The main reason for the superiority of CBCT over other imaging modalities is that CBCT provides images of high contrast and resolution for soft tissues and bone structures (Murphy et al., 2007). This helps in comparing images between the new and previous images in 3-dimensions with a high level of accuracy (Posiewnik and Piotrowski, 2019).

Although there are obvious benefits from applying the IGRT procedure, attention should be paid to the dose from the imaging procedure, as the imaging dose is delivered to a larger field surrounding the tumour.

1 Since multiple scans are performed for each patient during the radiotherapy treatment course, a daily CBCT
2 scan is acquired for some cases, the size-specific effective dose of those scans would be non-negligible, and
3 the dose level could be tens mSv. Effective dose is a dose quantity applied to a reference individual of a
4 standard size, so cannot be used to evaluate differences due to patient stature. Therefore, a modified version,
5 called the size-specific effective dose was used for this study (Martin et al., [2020](#); Martin et al., [2021](#)). For
6 instance, the size-specific effective dose from a single CBCT scan using the default chest or pelvis protocols
7 can vary over the ranges of 2.3 – 4.1 mSv and 4.6 – 9.9 mSv, respectively, depending on the size of the
8 patient (Martin and Abuhaimed, [2022](#); Abuhaimed and Martin, [2023](#)). Various recommendations and
9 approaches have been suggested to optimize the use of the IGRT procedure. The American Association of
10 Physicists in Medicine (AAPM), for example, has established two task groups, namely 75 and 180, from
11 which recommendations were provided (Murphy et al., [2007](#); Ding et al., [2018](#)), as well as the International
12 Commission on Radiological Protection (ICRP) publication 129 (ICRP, [2015](#)).

13 Two kV imaging systems of CBCT scans are commonly used for the IGRT procedure: (1) on-board
14 imaging system (OBI) by Varian, and (2) X-ray volumetric imaging system (XVI) by Elekta. The Varian
15 OBI system is widely implemented around the world (C. J. Martin et al., [2021](#)). Imaging doses resulting from
16 those systems are well documented in the literature and some studies have been reviewed by Alaei and Spezi
17 ([2015](#)). The use of the OBI system for the IGRT procedure is slightly different from that of conventional CT
18 scanners in that default scanning protocols with constant parameters such as kV, mAs, and size of the field
19 of view (FOV) are developed by Varian and applied for all patients regardless of the size. This differs from
20 the XVI system that is equipped with various collimator and filter cassettes, each of which gives a different
21 FOV size, and selection of the cassette depends on the patient size (Spezi et al., [2009](#)). Additionally, several
22 scanning parameters of conventional CT scans such as mAs and FOV size are variable and changeable based
23 on the patient size and the area of interest. This study aimed to investigate the use of patient-specific FOVs
24 for CBCT scans, and the impact of applying variable FOVs with three predefined sizes (Small, Medium, and
25 Large) on patient doses.

2. Materials and Methods

2.1. Simulations of X-ray beams

Organ and size-specific effective doses resulting from CBCT scans were estimated by using Monte Carlo (MC) simulations. The OBI system was simulated with BEAMnrc/EGSnrc user code based on geometrical and elemental composition information obtained from Varian (Rogers et al., 1995; Kawrakow I. et al., 2000; Walters et al., 2002). The MC model was validated and compared against the results of five different physical experiments. The development of the model and the validation works have been documented in detail in previous studies (Abuhaimed et al., 2015, 2014). In this study, three FOVs of different sizes were simulated at 120 kV using the body scanning protocol that is used for chest and pelvis CBCT scans V2.5 (Abuhaimed et al., 2018). Extension of the FOVs in the z-direction, i.e. along the patient's body, was set to (S = 18 cm), (M = 22 cm), and (L = 26 cm). The lateral size of the FOVs was kept constant at 49.4 cm for all cases, which was similar to the default size (Abuhaimed and Martin, 2018a). The supercomputing laboratory (SHAHEEN) at King Abdullah University of Science and Technology (KAUST) was utilized for running 1×10^{10} particles for each FOV using ISOURCE: 10 of the code (Parallel Circular Beam Incident from Side) (Rogers et al., 2023).

The electron transport cut-off (ECUT) of 0.516 MeV and photon (PCUT) of 0.001 MeV were set for all simulations. The directional bremsstrahlung splitting (DBS) technique with the split value set to 2×10^4 was utilized to optimize the efficiency of the simulation (Kawrakow et al., 2004; Mainegra-Hing and Kawrakow, 2006). The databases of the National Institute of Standards and Technology (NIST), Koch–Motz, and NIST XCOM were used for bremsstrahlung cross-sections, bremsstrahlung angular sampling, and photon cross-sections, respectively. The boundary crossing algorithm (EXACT) and electron step algorithm (PRESTA-II) were kept as default. The spin effect, the electron impact ionization, Rayleigh scattering, and atomic relaxations were turned on. X-ray beams for the various FOVs generated by the OBI system were simulated and saved in phase space files just below the exit window of the X-ray tube. These files were considered as x-ray sources for the tube, with which doses could be assessed. Thus, the files were transferred to another user code to estimate organ and size-specific effective doses.

2.2. Estimation of organ and size-specific effective doses

Organ and size-specific effective doses were assessed using a voxel-based user code, namely DOSXYZnrc/EGSnrc (Kawrakow and Walters, 2006; Walters et al., 2023). The phase space files obtained with BEAMnrc code were input as x-ray sources of the OBI system of various FOV sizes, and two regions of interest were investigated, chest and pelvis. Organ and size-specific effective doses were estimated on a library of mesh-type computational human adult phantoms developed by the National Cancer Institute (NCI) and the University of Florida (Geyer et al., 2014; NCI, 2023). The library contains 193 adult phantoms that cover the whole adult population with a wide range of heights and weights. To investigate the impact of the FOV size, on the patient doses, the phantoms were grouped based on the height as summarized in Table 1. The weight of the phantom was increased from the smallest to the largest weight in intervals of 5 kg for each group listed in the table. The phantom library was created based on thousands of images from patients who had undergone CT scans, thus the mesh-type phantom can precisely define the patient body.

Table 1: A summary of specifications of the adult phantoms involved in this study.

Male			Female		
Height (cm)	Weight (kg)	No. of Phantoms	Height (cm)	Weight (kg)	No. of Phantoms
160	50 – 95	10	150	40 – 95	12
165	50 – 110	13	155	40 – 115	16
170	50 – 125	16	160	45 – 125	17
175	55 – 135	17	165	45 – 135	19
180	55 – 140	18	170	50 – 125	16
185	65 – 135	15	175	60 – 120	13
190	75 – 125	11			
Total		100			93

The elemental compositions and densities of voxels of the phantoms were defined in DOSXYZnrc using PEGS4 code as obtained from the NCI. The initial step required converting the format of the phantoms to that of DOSXYZnrc using a MATLAB code developed for this purpose. This step involved converting the format of phantom matrixes of large sizes ranging from (267 x 223 x 561 voxels) up to (345 x 373 x 652 voxels). To investigate the influence of the scan FOV on the patient dose, three simulations were run for each phantom on the chest region and another three on the pelvis region, each of which used different FOVs. Therefore, six simulations were required for each phantom, which resulted in a total of 1,158 simulations conducted with DOSXYZnrc. The middle of the lungs and urinary bladder were considered as centres of the chest and pelvis scans, respectively, as illustrated in Figure 1.

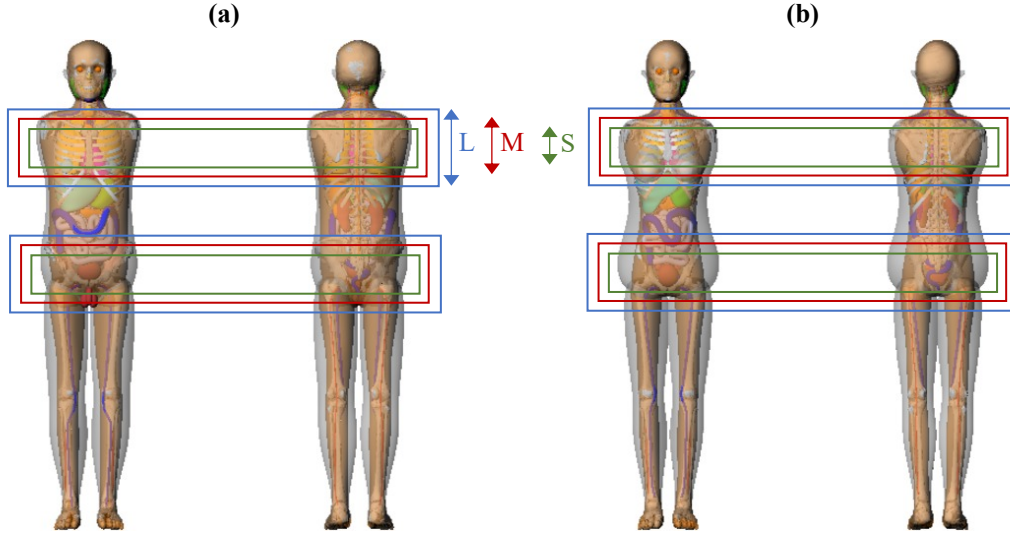


Figure 1: An illustration of the scan regions covered by the different FOVs on the (a) male and (b) female phantoms. The small (S) FOV covered 18 cm along the patient's body, whereas 24 cm and 26 cm were used for the medium (M) and the large (L) FOVs.

All particles stored in the phase space files were run in DOSXYZnrc, and the sources were tracked through a full 360 degree rotation to represent the CBCT acquisition using the ISOURCE: 8 of the code (Phase-Space Source Incident from Multiple Directions) (Walters et al., 2023). Source to isocentre distance (SAD) was set to 100 cm for all the simulations, where the isocentre was the centre of the FOV, over which the source was rotated. The default boundary crossing algorithm (PRESTA-I) was utilized. The same cross-sectional databases and Monte Carlo parameters used in BEAMnrc simulations were also applied to DOSXYZnrc simulations. The results of each run were stored in the (3ddose) format of the user code. A MATLAB code was written to analyze the dose information of each phantom by determining and extracting voxels of each tissue and organ, which were assigned with specific IDs. The mean dose was calculated for each tissue and organ, and the size-specific effective dose for each scan was assessed using the radiation and tissue weighting factors from ICRP Publication 103 (ICRP, 2007). The method used for calculations of organ and size-specific effective doses were reported in detail in previous studies (Martin et al., 2016; Abuhaimed et al., 2018; Abuhaimed and Martin, 2018b). All simulations carried out with DOSXYZnrc were also run within the supercomputer (SHAHEEN).

3. Results

3.1. Assessment of scan doses

Doses to organs that were fully or partially inside the FOVs were considered as organs that were outside the scan region would only receive a very low dose, mainly from scattered radiations associated with a high uncertainty. Therefore, doses to the red bone marrow (RBM), lung, stomach, breast, heart, oesophagus, liver, and thyroid were selected for the chest scan, and are presented in [Figure 2](#) for the male and female phantoms. [Figure 3](#) shows doses to the RBM, colon, small intestine, prostate, uterus, urinary bladder, and gonads (testes and ovaries) resulting from the pelvis scan for all the phantoms. Size-specific effective doses resulting from both scan protocols are also presented in [Figure 4](#).

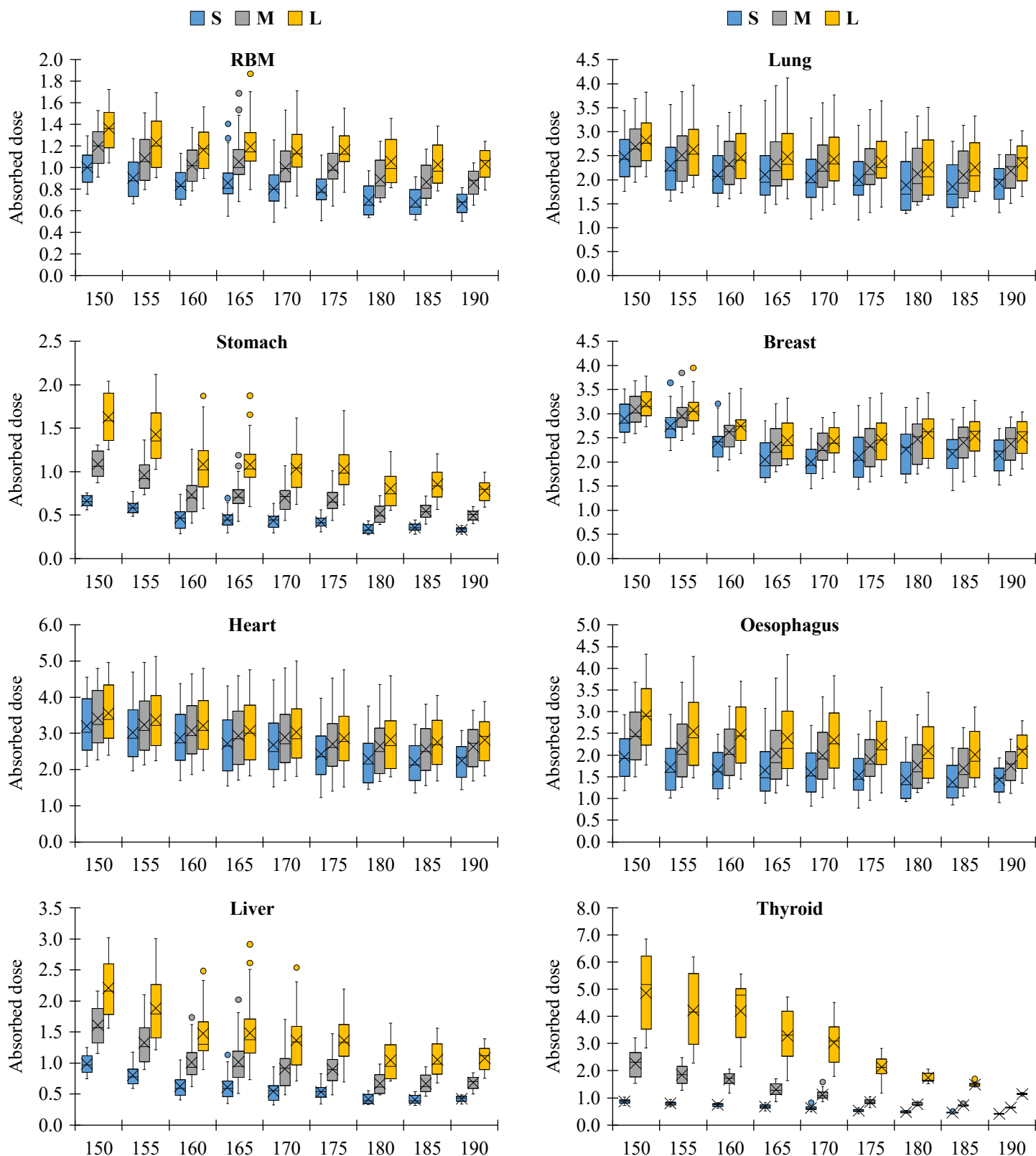
As can be seen from the figures, generally, organ doses are raised with the increase in the FOV size for all organs and decreased with the phantom height. Mean doses to organs that were fully inside the FOV were comparable for the male and female for all scans. For instance, the mean doses to lung and heart of the chest scan were 2.2 and 2.8 mGy/100 mAs for the male, and 2.4 and 2.9 mGy/100 mAs for the female, respectively. In addition, the mean dose to the urinary bladder of the pelvis scan for both genders was also comparable, 2.9 and 2.7 mGy/100 mAs for the male and female, respectively.

However, doses to organs that were covered partially at the boundaries of the FOVs were affected by the FOV size remarkably. A significant influence was noticed for doses to the thyroid of the chest scan for both genders and dose to the testes of the male pelvis scan. Besides the FOV influence, thyroid doses of the chest scan were also affected significantly by the phantom height. The doses started from 6.8 mGy/100 mAs at the shortest phantom and decreased gradually reaching 0.4 mGy/100 mAs for the higher phantoms. The dose to the testes was mainly influenced by the phantom height, where doses of all phantoms were in the range of 0.8 – 7.0 mGy/100 mAs. The dose variations of testes were small for shorter phantoms (160, 165, and 170 cm), over which the different FOVs fully or partially covered the testes. Therefore, mean doses for these heights were comparable, 5.8, 5.8, and 5.6 mGy/100 mAs, respectively. This was followed by a significant dose reduction for the taller phantoms, particularly the dose resulting from the small FOV because testes had moved outside the FOV. As a result, mean doses to the testes using the small and medium FOVs fell to 1.0 and 3.2 mGy/100 mAs, respectively, while a minimal change was found for doses with the large

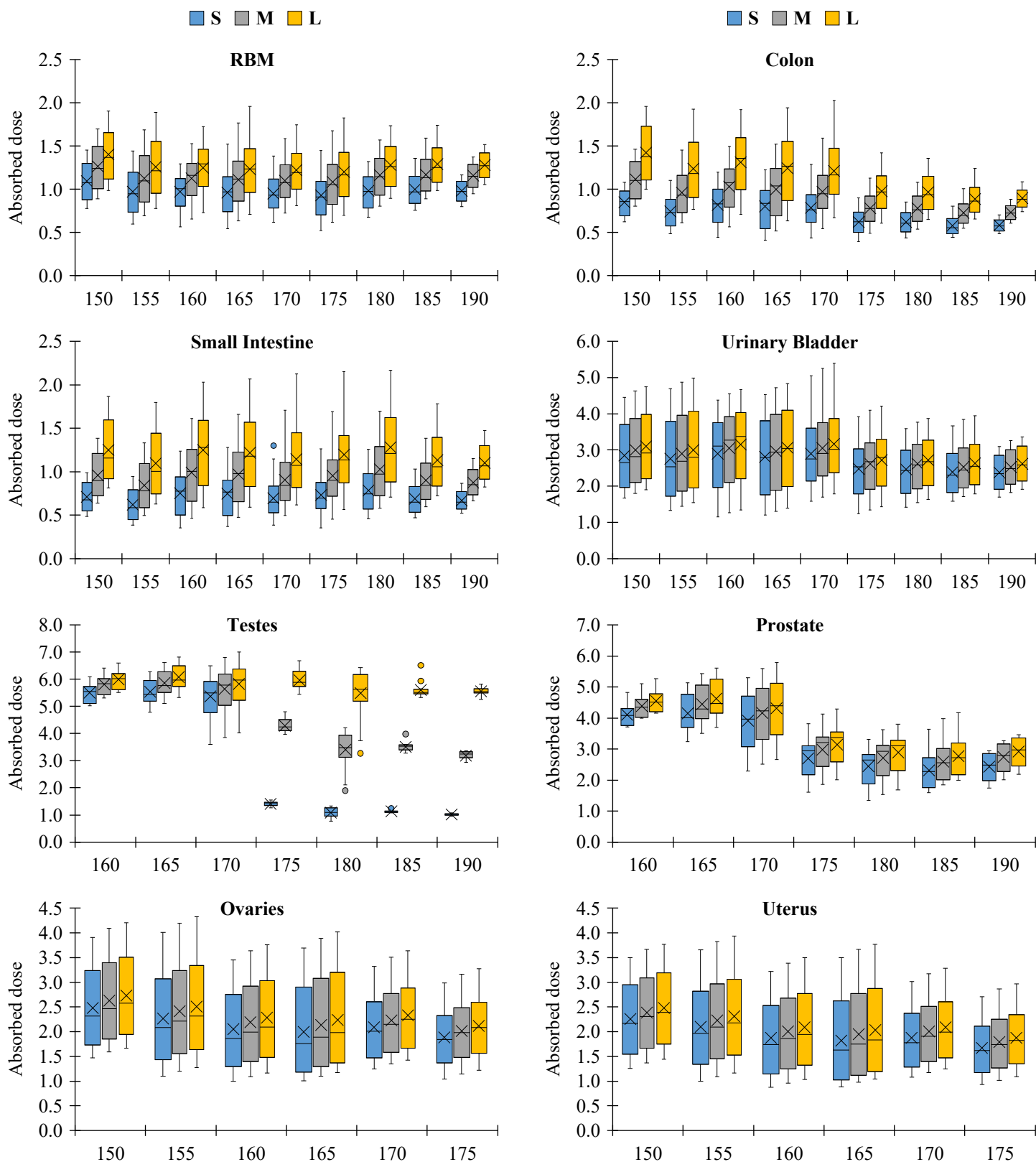
1 FOV for all heights. Doses to the stomach, liver, oesophagus of the chest scan, and doses to the colon and
2 small intestine of the pelvis scan were also affected, yet the dose variations were smaller than those of the
3 thyroid and testes.

4 As a result of those variations, size-specific effective doses varied widely for the scan protocols
5 investigated, ranging from 0.5 to 2.2 mSv/100 mAs for the chest and from 0.3 to 1.3 mSv/100 mAs for the
6 pelvis. The large influence and variation of the size-specific effective doses seen for the pelvis scan (Figure
7 5) is due to the doses to the testes (Figure 4), which are a relatively small organ, on the edge of the scan field,
8 that are excluded from the irradiation field with the small field size, and have a tissue weighting factor of
9 0.08 (ICRP, 2007). A similar effect is seen for the thyroid in the chest scans (Figure 3 and 5), which is similar
10 in size to the reductions in field size that are made, and having a tissue weighting factor of 0.04 of the size-
11 specific effective doses for smaller patients.

12



1 **Figure 2:** Box and whisker plots of organ doses (mGy/100 mAs) of the chest scan using various FOVs on
2 all the phantoms ([Table 1](#)).



1 **Figure 3:** Box and whisker plots of organ doses (mGy/100 mAs) of the pelvis scan using various FOVs on
2 all the phantoms (Table 1).

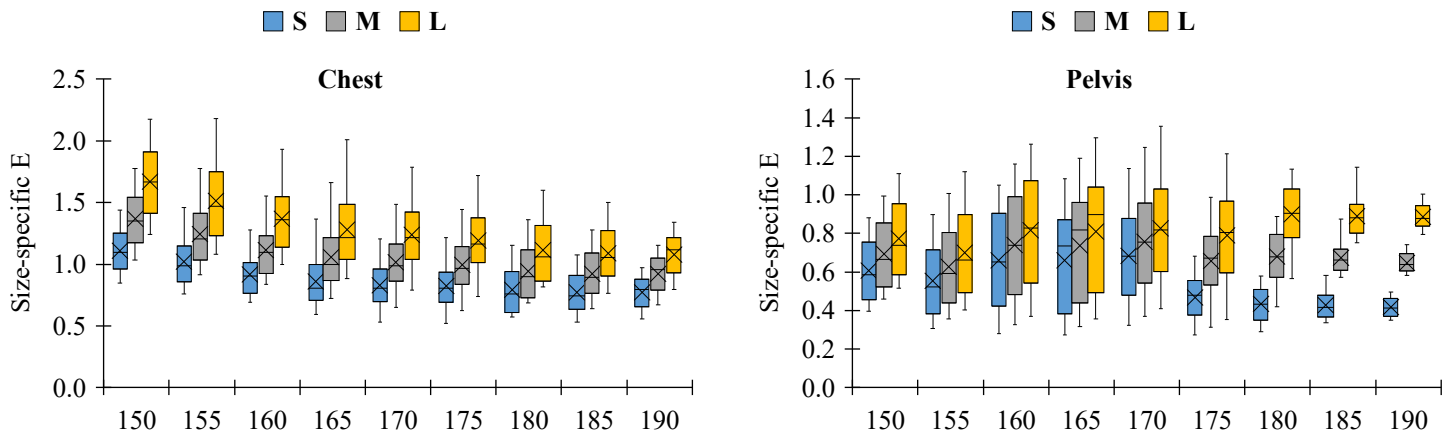


Figure 4: Box and whisker plots of size-specific effective doses (mSv/100 mAs) of the chest and pelvis scans using various FOVs on all the phantoms (Table 1).

3.2. Influence of the FOV size

As would be expected, organ and size-specific effective doses were affected by the size of the FOV as well as the height of the phantom. Figure 5 presents the average percentage differences (PDs) in organ and size-specific effective doses resulting from the increase in FOV of the chest and pelvis scans by 4 cm from the small (18 cm) to the medium (22 cm) sizes, and from the medium to the large (26 cm) FOV. Generally, the average PDs of organs, that were inside the FOVs, were less than 12% for both scan protocols, but large differences were found for the other organs. The average PDs of organs at boundaries of the FOVs ranged from 15% to 137% for the chest scan, and from 11% to 130% for the pelvis. The average PDs of size-specific effective dose were comparable for both scan protocols and were within 27%.

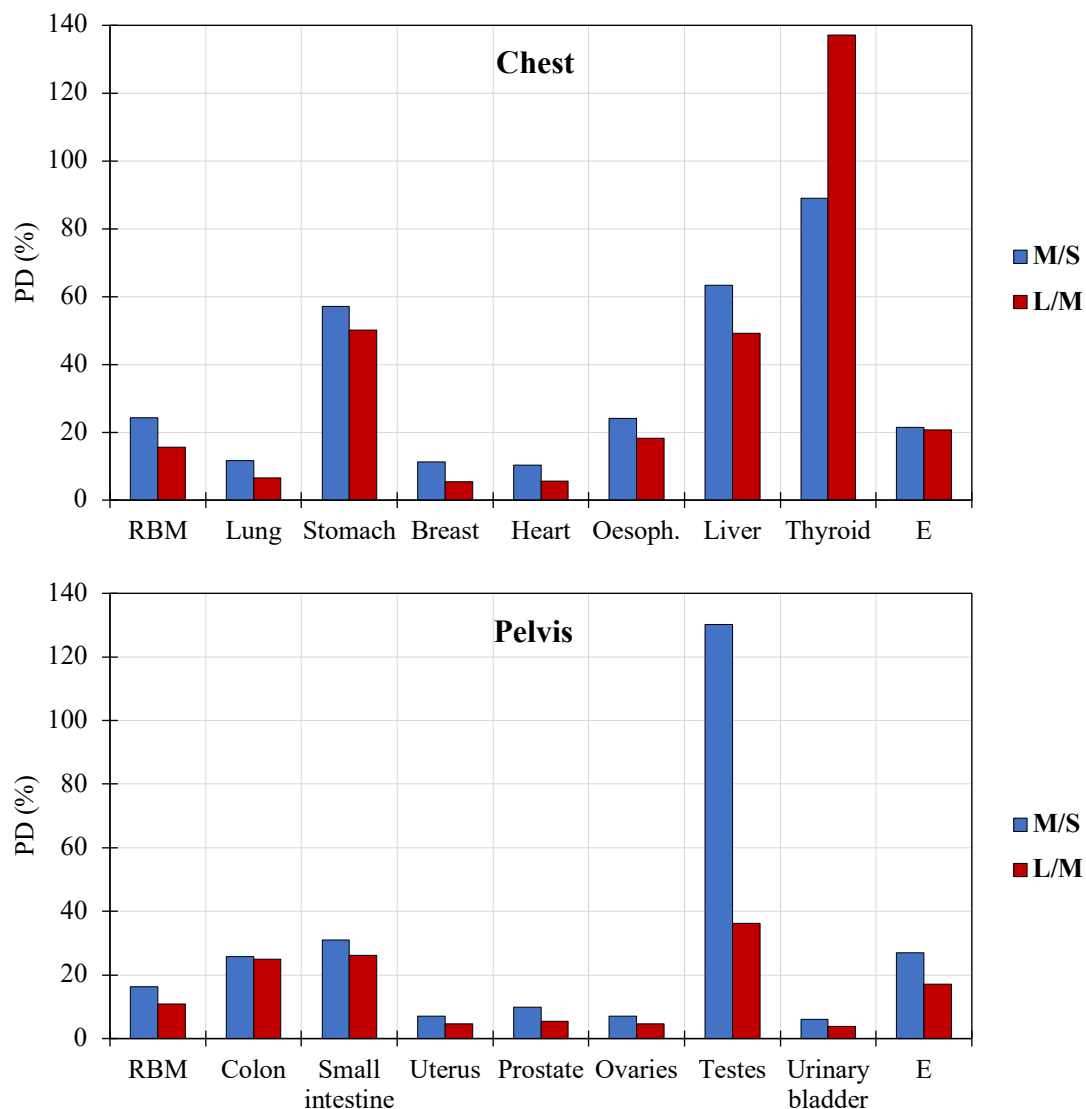


Figure 5: Average percentage differences (PDs) in organ and size-specific effective dose between using the small (S) and medium (M) FOVs, and between the medium and large (L) FOVs for the chest and pelvis scans.

Compared to the small FOV, the use of the medium FOV for both scan protocols led to dose increases in the range of 4 – 29% for organs that were fully covered by the FOV. However, the PD values were significantly higher for organs that were at the boundaries of the FOV. The medium FOV raised thyroid and testes doses by up to 215% and 233% for the chest and pelvis scans, respectively. On the other hand, the dose increase was slightly lower when the FOV was extended to the large size as compared to the medium size, except for the thyroid. Organ doses were increased by 2 – 22% for the chest and pelvis scans, but the highest

PD in doses for the thyroid and testes reached 210% and 79%, respectively, when the large FOV was used. Accordingly, size-specific effective dose values were affected by the FOV size, where the doses of the chest scan were increased by 18 to 24% between the small and medium FOVs, and by 16 to 26% between the medium and large FOVs. However, variations in the size-specific effective dose for the pelvis scan were higher than that for the chest due to testes doses.

3.3. Influence of the patient size

Only the influence of height and field size on organ and size-specific effective doses have been investigated in detail in this study, but adjustments would also be required to weight for complete optimization of the protocols. Organ and size-specific effective doses were found to be affected significantly by the weight for some phantoms that had the same height and exposed to the same field size. Generally, the weight influence decreased with increasing the phantom height, and this can be seen in the box plots shown in [Figure 2 – 4](#). For the chest scan, the largest influence for the weight was found in the thyroid dose for female phantoms with height of 150 cm and 155 cm that were scanned by the large FOV ([Figure 2](#)). The dose difference between the smallest and largest sizes was found to be 4 mGy/100 mAs. Similarly, the weight had an impact on doses to urinary bladder, ovaries, and uterus of the pelvis scan ([Figure 3](#)). This is in agreement with the influence reported in a recent study (Fetin et al., [2022](#)). Doses to these organs were influenced by the weight in the range of 2 – 4 mGy/100 mAs. Subsequently, variations of size-specific effective doses of the phantoms that had the same height reached 1 mSv/100 mAs for the both scan protocols ([Figure 4](#)).

4. Discussion

It has been suggested that the patient dose resulting from CBCT scans can be roughly estimated assuming that the dose variations between patients of various sizes are relatively small (Ding et al., [2018](#)). This method was defined as a nonpatient-specific estimation approach, and based on using tabulated values for each scan protocol. Doses to some organs presented in the previous sections are compared with those published in the AAPM TG 180 report as typical dose values and shown in [Table 2](#). It can be seen that, generally, the results of this study are in agreement with those of the AAPM. However, some discrepancies were found for a few organs such as the lung and urinary bladder. Several reasons might be behind such differences: (1) the dose values presented in the AAPM report were based on a study conducted with scan

protocols of V1.4, which were slightly different from the updated V2.5 protocols used in this study (Nelson and Ding, 2014). One of the main differences between the versions is the extension of the FOV in the z-direction, i.e. the scan beam width. A beam width of 19.8 cm is used for V1.4 protocols, whereas 21.4 cm is applied for V2.5 (Abuhaimeed et al., 2018). It should be noted that doses in this study presented in Table 2 were based on those acquired with the medium FOV. (2) Doses presented in the AAPM report were based on a small number of patients, 8 for the chest and 5 for the pelvis (Nelson and Ding, 2014). In contrast, a total of 193 phantoms with a wide range of sizes were used in this study ranging from as small as 150 cm and 40 kg up to 190 cm and 125 kg. (3) The lung and urinary bladder were placed at the centre of all simulations for the chest and pelvis scans, respectively. This means that those organs as well as the heart and prostate were fully inside the FOVs of the different sizes. These reasons may explain the higher doses for these organs as compared to the AAPM values.

Table 2: Comparisons between doses of CBCT scans published in the AAPM TG – 180 report (Ding et al., 2018) and the results of this study.

Chest			Pelvis		
Organ	AAPM	This study	Organ	AAPM	This study
	mGy / 100 mAs			mGy / 100 mAs	
Lungs	1.1 – 2.3	1.3 – 4.0	Urinary bladder	1.3 – 2.1	1.3 – 5.3
Stomach	1.1 – 2.2	0.4 – 1.4	Small intestine	1.5 – 1.8	0.5 – 1.7
Heart	1.2 – 2.1	1.4 – 5.0	Prostate	1.1 – 1.7	1.5 – 5.6
Oesophagus	1.1 – 2.3	1.0 – 3.8			
Liver	1.2 – 1.9	0.5 – 2.2			

The results presented in Figure 1 – 4, clearly show that the size of the FOV as well as the patient size would play a significant role in doses of CBCT scans. Different approaches were investigated in various studies published in the literature to avoid delivering unnecessary doses to the patients and acquire an acceptable image quality at relatively lower doses. Inclusion of the kV imaging doses into the total treatment dose by using the available treatment planning systems was one of the earliest methods suggested to account for doses of CBCT scans (Alaei et al., 2010, 2014; Alaei and Spezi, 2012; Dzierma et al., 2014; Albarakati et al., 2022). This would be useful for maintaining doses to organs at risk below the tolerance doses. However, this method was associated with some challenges; for example, the treatment planning systems are developed, tested, and calibrated for the treatment energies in the MV range. The use of those systems exhibits limitations in the assessment of imaging doses at the low kV energies, with which imaging scans are acquired. Therefore, a modification to the treatment planning system followed by beam commissioning is required to account for

1 the imaging dose, and this option is unavailable currently. The development of model-based dose calculation
2 algorithms for kV energies was also proposed, not only for doses from CBCT scans but also for radiographs
3 (Pawlowski and Ding, 2014). Studies showed that large uncertainties are associated with this approach
4 because current models do not account for the photoelectric effect. This results in an underestimation of the
5 dose to bone structures reaching 300% (Ding et al., 2018).

6 Some studies have recommended the establishment of several scanning protocols of variable FOVs that
7 are patient-specific protocols. It has been suggested that CTDI_{vol} values of CT scans acquired for patients
8 with simulators for treatment planning are used to divide patients into several groups, each of which is
9 scanned with a different scan protocol (Ordóñez-Sanz et al., 2021). This approach has been shown to optimize
10 doses for pelvis CBCT scans remarkably. It contributed to dose reductions ranging from 19 – 83% for all
11 patient sizes, except for obese patients, where doses were increased by 39%. In addition, Khan et al., 2022,
12 have suggested the establishment of several patient-size-specific CBCT scanning protocols for each scan
13 region, which cover the head and neck, thorax, and abdomen and pelvis regions. The average patient diameter
14 of the scan region was considered to select the appropriate protocol to scan each patient. It has been found
15 that this approach led to dose reductions for 52% of the patients, while doses for 37% of the patients were
16 similar to the default protocols, and increased for 12% of the patients.

17 A dynamic blade system that can be implemented into the integrated kV imaging system was developed
18 by Parsons and Robar (2016). This allows changing the FOV size of CBCT scans dynamically during
19 acquisition of the scan by tracking the area of interest. The results showed that this system could result in a
20 15 – 80% dose reduction for CBCT scans. Moreover, the implementation of the tube current modulation
21 technique into the kV system to be used like that applied for conventional CT was investigated (Stankovic et
22 al., 2021). The use of the modulation technique that modulates the current based on the patient body during
23 the scan acquisition in a manner similar to that used for conventional CT was found to improve the quality
24 of CBCT images by up to 30% with relatively lower doses. Recently, utilizing a new acrylic-lead shielding
25 device has been suggested to reduce doses of organs that are near to boundaries of the scan FOV (Shimizu et
26 al., 2022). The study showed that reductions of 45% and 47% in doses of the eye lens and thyroid,
27 respectively, could be obtained by using this device. Despite these studies investigating practical approaches

to reduce patient doses, implementation of the systems may not be easily accomplished in the clinic, and such improvements could be performed by the manufactures.

Figure 6 shows comparisons between dose-volume histograms for the thyroid resulting from the different FOVs for the shortest male (160 cm and 50 kg) and female (140 cm and 40 kg) phantoms. The use of the small FOV could be sufficient for short patients, which helps in minimizing the exposure level. The main thyroid dose for the shortest male and female phantoms were 0.8 and 1.0 mGy/100 mAs, respectively, when the small FOV was used. These doses were increased to 2.0 and 3.2 mGy/100 mAs, for the medium FOV, and to 5.1 and 6.8 FOV mGy/100 mAs for the large, respectively. This means that a significant dose reduction could be achieved if the small FOV is sufficient for the short patients to meet their clinical needs. The results suggest that the development of several scanning protocols for each scan region should be considered to effectively reduce doses of CBCT scans, and the setting for the field size in the protocol should be based on the patient's height and weight. The height could be used to determine extension of the FOV, whereas the scan paraments such as the tube potential (kVp) and current (mAs), could be based on the weight. This approach may be more practical and can be easily implemented in the clinic to overcome the issue of using a fixed FOV for all patients of different sizes.

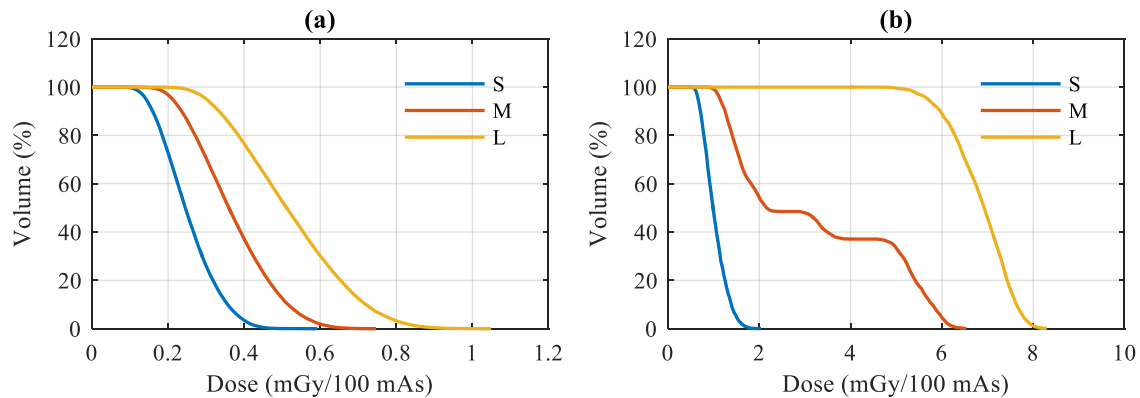


Figure 6: Comparisons of dose-volume histograms (DVHs) of thyroid doses using the S, M, and L FOVs for (a) the male (160 cm and 50 kg), and (b) the female (150 cm and 40 kg).

Although the Varian OBI is the most common imaging system used for CBCT scans of the IGRT procedures, the use of a single system in this study is considered one of the limitations. Moreover, this study was based on Monte Carlo simulations, thus some statistical uncertainties are expected to occur. The results

1 were acquired on adult computational human phantoms, which might be slightly different from doses
2 assessed on CT images of patients.

3 **5. Conclusions**

4 Monte Carlo simulations were used to estimate organ and size-specific effective doses resulting from
5 chest and pelvis CBCT scans on adult computational human phantoms using three FOVs of different sizes.
6 The results show that the imaging dose was affected by the size of the FOV applied for the scan. A significant
7 influence was found for organs that were covered partially by the FOV or located near the scan area,
8 particularly small organs such as the thyroid and testes. Doses to organs that were inside the FOVs did not
9 change substantially with the FOV size, where all average PDs were less than 12%. However, doses to the
10 thyroid in the chest scan for both genders and testes for the male pelvis scan were changed notably with the
11 FOV size. The average PDs of these organs reached as high as 130%. Since changing the field size option is
12 unavailable for the OBI system, it can be suggested that various patient-specific scanning protocols could be
13 developed for each scan area. The patient's height might be considered as the main factor upon which, the
14 selection of the patient-specific field size in the protocol is based as well as the weight. This approach should
15 assist in optimizing the imaging doses.

16 **Acknowledgments**

17 The authors would like to extend their sincere appreciation to the Researchers supporting program for
18 funding this work under Researchers Supporting Project number (RSPD2023R780), King Saud University,
19 P.O. Box 145111, Riyadh 4545, Saudi Arabia. The authors would also like to thank the National Cancer
20 Institute of the National Institutes of Health (NIH) of the US for sharing the phantoms library used in this
21 study, and the supercomputing lab at King Abdullah University of Science and Technology (KAUST) for
22 their permission of performing all Monte Carlo simulations on the supercomputer (SHAHEEN).

23

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1 **Statements and Declarations**

2 **Funding**

3 This work was supported by the researchers supporting program at King Saud University, Riyadh, Saudi
4 Arabia with a grant number of RSPD2023R780.

5
6 **Competing Interests**

7 The authors have no relevant financial or non-financial interests to disclose.

8
9 **Author Contributions**

10 All authors contributed to the study conception. Material preparation, design, data collection and
11 analysis were performed by Abdullah Abuhaimed and Huda Mujammami. The first draft of the
12 manuscript was written by Abdullah Abuhaimed and all authors commented on previous versions of
13 the manuscript. All authors read and approved the final manuscript. No human or animal data were
14 included in the study.